SPACE COLONY — PROJECT PLAN

Team:

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A group-developed Android app where players manage a Space Colony crew. Players can recruit five specializations (Pilot, Engineer, Medic, Scientist, and Soldier), train them in a Simulator to gain experience, and send two-member squads on cooperative turn-based missions against system-generated threats. Defeated crew members are permanently removed from the program.

Core Features:

Must-Have:

🎯 Create Crew Members (name + specialization selection)

🎯 Simulator (+1 XP per session → +1 skill power)

🎯 Mission Control (cooperative turn-based missions vs system-generated threat)

🎯 Quarters (health fully restored when returned home, XP retained)

🎯 Stats Tracking (skill/resilience/energy/XP)

Nice-to-Have Features (If Time Allows):

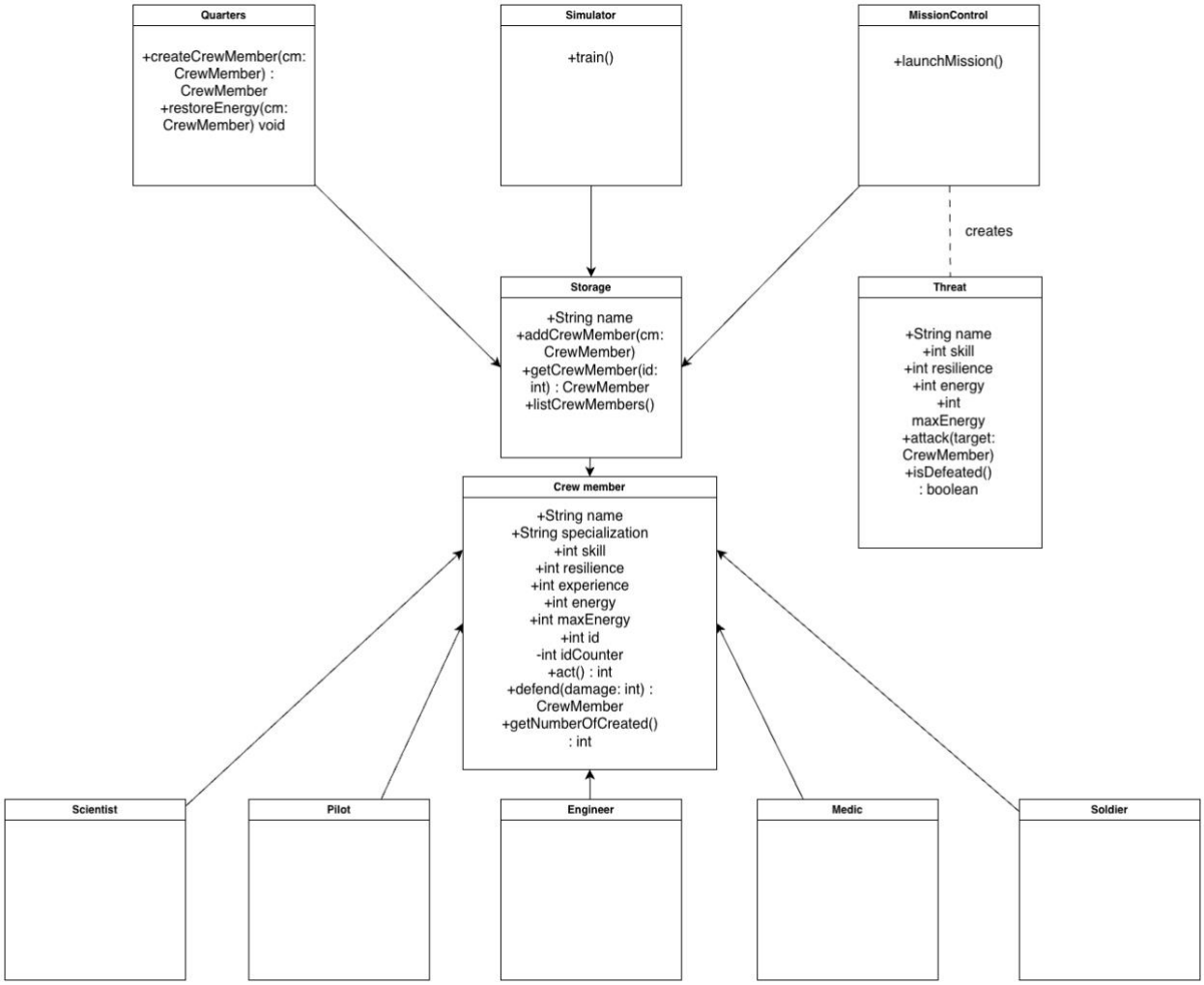
⚔ Crew Images (unique image per specialization)

⚔ RecyclerView (crew members listed using RecyclerView component)

⚔ Randomized Damage (attacks vary slightly with Math.random())

⚔ SPACE COLONY CLASS ARCHITECTURE ⚔

Training, Missions, and Beyond



User Interface Description

6 Core Views/Screens:

1. Home Screen: Function: Shows summary of crew members in each location (Quarters, Simulator, Mission Control). **Actions:** Buttons to "Recruit New" or navigate to Simulator/Mission Control. UML Note: HomeView -- ColonyOverview.

2. Recruit Crew Member Screen: Function: Input name + select specialization (with stats preview). UML Note: RecruitView --> CrewMemberFactory.

3. Quarters Screen: Function: List of crew members at home with their stats. **Actions:** Buttons to "Move to Simulator" and "Move to Mission Control". UML Note: QuartersView ..> QuartersController.

4. Simulator Screen: Function: Select crew member → "Train" button (+1 XP). **Visual:** Progress bar showing XP growth. UML Note: SimulatorView ..> SimulatorController.

5. Mission Control Screen: Function: Select 2 crew members → "Launch Mission" → Animated mission log. UML Note: MissionView --> MissionControl.

6. Statistics Screen: Function: Show missions/victories, training history per crew member. UML Note: StatsView -- StatsController.

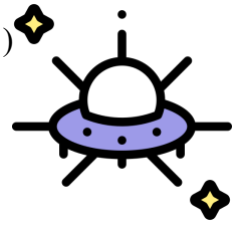
Bonus:

7. Mission Result Screen:

Function: Display mission outcome (threat neutralized or mission failed) + XP earned by surviving crew members.

Planned Bonus Features:

⚔ Crew Images — Different crew specializations visualized with unique images (+1)

⚔ RecyclerView — Crew members listed using RecyclerView component (+1)

⚔ Randomness in Missions — Attack damage varies slightly using Math.random() (+1)